

Technology Stack

Date	15 July 2026
Team ID	SWTID-2026-2749
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client Side	HTML5, CSS3, JavaScrip	Developed a responsive and user-friendly interface for farmers

2	Backend / Server-Side	Python, Flask	Handles prediction requests and connects the web interface with the Machine Learning model.
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	CSV Dataset (Crop Recommendation Dataset)	Stores crop training data used for Machine Learning model training.
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	Local Flask Server (Can later be deployed on Render/AWS)	Used for development and testing with future cloud deployment support.
5	Version Control & CI/CD	Git, GitHub	Maintains version history and enables collaboration and code backup.

	<i>(e.g., GitHub, GitLab, Jenkins)</i>		
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn	Used for Machine Learning, data preprocessing, analysis, visualization, and prediction.